

Article

Evaluating Livability Perceptions: Indicators to Evaluate Livability of a University Campus

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Abstract: Livability indicators and assessments are critical to promoting campus sustainability and livability. Despite the various indicator frameworks and local initiatives aimed at attaining sustainability, campus livability assessments in the Middle East are scarce. An extensive literature search was used in this paper to obtain the best livability indicators, and the Delphi method was used to reach consensus using a panel of experts in the fields of sustainability and livability. The determining indicators were identified and categorized according to Environmental Quality, Social and Cultural Elements, Accessibility, Amenities and Services, Safety, Housing, and Other Services. A questionnaire using both qualitative and quantitative questions was used, taking into consideration the perceptions of students, faculty, and staff residing at the American University of Sharjah (AUS) campus in the United Arab Emirates (UAE). A score of 87/100 was obtained in this study, indicating that the AUS can be considered a highly livable campus. Responses showed that social and cultural elements, such as social gatherings, availability of restaurants, and children's activities, were of utmost importance to the respondents. Additionally, services on campus, accessibility, and quality of housing played a major role in respondents' choices. The index obtained in this study serves as a reference/benchmark for similar and comparative studies of campus livability, and can be adapted to similar scenarios. Moreover, this study opens the doors for future studies to gain deeper insights into the required steps to evaluate the livability of a campus, in order to facilitate future transformation of a traditional campus into a livable campus.

Keywords: livable campus; Delphi method; Relative Importance Index (RII); livability indicators; sustainability



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1. Introduction

In spite of its presence in academic and professional readings, the term livability is considered subjective and is used differently by diverse groups, depending on the context [1,2]. Thus, no consensus on how to define livability and its related factors has been reached, despite the growing number of academics and professionals who are involved in livability matters. In essence, a clear understanding of livability is needed [1–4]. Livability is a significant and essential set of factors or conditions that characterize quality of life. It refers to numerous constructed interpretations regarding the quality of life in any human living setting. This concept aims to enhance quality and integrity of human life; hence, a fundamental objective in urban design and planning is the achievement of urban livability [5]. Some of the key factors in interpreting the concept of livability includes peoples' lifestyles, cultural norms, and daily activities [6,7]. Therefore, a local and contextual interpretation of livability is a must.

1.1. Defining Livability

The concept of 'livability' lacks any specific and universally accepted definition due to its complex and multi-dimensional nature. The variety of available studies on livability

embrace similar notions, such as sustainability, quality of life, well-being, health of communities, and the ‘character’ of a place [8,9]. Mohit et al. argue that [10] the main issue with the concept of livability is that scholars form definitions that are suitable for their specific research areas and not for the general idea. The substantial body of literature shows different interpretations of what livability is depending on the stakeholder’s perspective, encompassing a broad range of factors and activities. For example, livability can be defined as ‘suitability for human living’ and ‘the standard of living or general well-being of the population in an area’ [11]. Likewise, urban livability is directly related to individuals’ satisfaction with performing their daily activities in familiar settings [8]. Other definitions tend to emphasize or focus more on the conditions of livability. Shabanzadeh et al. defined a livable community as a community that is not only affordable, but that includes community support, services, proper housing, and adequate mobility, and whose facilities enhance individual autonomy and participation of residents in civil and social life [2].

There are other unique but enlightening definitions of livability in the literature. An example is Stanislav et al., who defined livability as ‘a set of physical conditions that integrate the natural and built environments and create safety and comfort, include engaging facades, and offer easy access to services and transit’ [12]. This definition integrates the social facets and amenities of the built environment, and also the physical facets that contribute to livability. From another viewpoint, Pacione [13] drew a link between livability and behavior, stating that ‘livability is a quality that is not an attribute inherent in the environment but is behavior-related function of the interaction between environmental characteristics and personal characteristics’. However, some definitions of livability might differ from time to time, as well as from one culture to another, as the concept of livability is relative ‘whose precise meaning depends on the place, time, and purpose of the assessment and on the value system of the assessor’ [13]. It is noted from the literature and definitions that many academics identify ‘livability’ as a subjective concept and term it using different jargon, including the terms ‘vitality’, ‘liveliness’, and ‘sense of belonging’. This study argues that all the expressions and terms mentioned above are qualities that a livable place must have, rather than synonyms for ‘livability’. In this study, the concept of ‘livability’ is defined as the ability of a living space to promote the well-being or quality of life of its residents [14].

From the examples of the definitions of livability cited above, this study has made some noteworthy observations. First, livability is a fluid, multifaceted, and broad concept with no precise or universally agreed-upon definition. Interestingly, there is nowhere in the literature where this is judged to be a negative or positive mark of livability discourse or research. An obvious problem with ‘livability’ is that it means something different to different people. Moreover, definitional perspectives and differences notwithstanding, a common denominator of all definitions is their emphasis on the wellbeing of people or society as the core of livability. Finally, ‘livability’ has become a popular term in planning, design, and engineering circles, yet there continues to be a lack of clear consensus about what livability actually means, let alone on how to measure it and how to achieve it. Having discussed the concept of livability in general, the following section discusses the concept of ‘livability’ versus ‘sustainability’.

1.2. Livability vs. Sustainability

As stated above, although livability is an important concept in planning, design, and engineering circles, there is a lack of clear consensus about its meaning, as well as on how to measure and achieve it. The concept of ‘livability’ has emerged alongside ‘sustainability’ as a buzzword in the public discourse [15]. According to Gough [16], it is generally assumed that people must have a right to both ‘sustainable’ as well as ‘livable’ communities, which prompts planners and decision-makers to consider how to address the needs of both the present and future residents. Nevertheless, the theoretical nexuses between livability and sustainability are not fully understood, making it difficult to agree on and evaluate policies to promote these ideals [17]. The World Commission on Environment and

Development (WCED) crafted in 1987 a definition for sustainable development, also known as the Brundtland Report, that has been acknowledged by the scientific discourse [18–20]. According to this report, sustainable development ‘is development that meets the needs of the present without jeopardizing the ability of future generations to meet their own needs’ [21]. The main contribution of the concept of sustainable development lies in its concrete consideration of future development [22].

Livability, in contrast to sustainability, brings an obligatory pragmatism to the philosophical concepts of sustainability. Livability emphasizes the ‘here’ and ‘now’, focused on immediate and tangible conditions and interventions, whereas sustainability emphasizes the ‘elsewhere’ and the ‘future’ [23,24]. This is what makes livability a fluid concept that changes according to the values and context of the community [12,15,16]. Therefore, livability is interpreted as more achievable than sustainability [15]. The terms livability and sustainability are complementary rather than antagonistic; ‘livability’ refers to a set of behaviors that people take in the context of the present, which, over time, lead to ‘sustainability’ in the long term [25]. This visualization is evidently illustrated in Figure 1, which shows that livability is a subset of sustainability, i.e., dependent on the triple bottom line (TBL) of the sustainability framework. Therefore, it cannot be considered as a stand-alone independent aspect, and no facet of livability conflicts with sustainability outcomes. Abdel-Hadi et al. [26] claim that livability studies only go beyond sustainability studies if human and social factors are taken more into account. The model proposed by Abdel-Hadi et al. [26] positions the narrower and more direct view of livability within the broader global and longer-term view of sustainability.

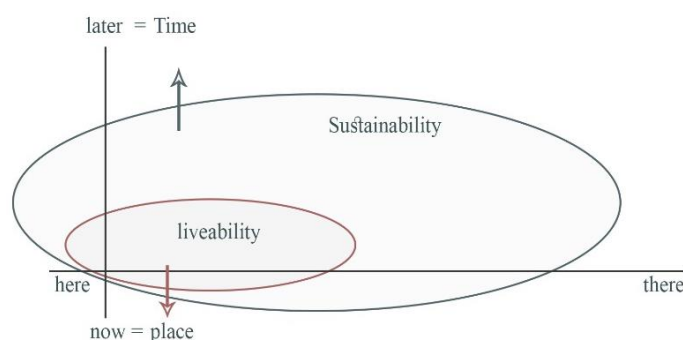


Figure 1. ‘Livability’ Vs. ‘Sustainability’.

There is widespread agreement that a livable community is one that offers choice and diversity of amenities to individuals who live and work in the community, despite the scholarly recognition that livability is a subjective and relative concept [27]. Therefore, investigating the impact on livability of amenities such as transportation, walkability, and access to open space allows for comparison among different societies in terms of their ‘livability’ [16,28–30]. Understanding how livability complements sustainability will assist policymakers to connect what residents need in the present with longer-term needs linked to sustainability. Although the noted differences between sustainability and livability indicate competing prescriptions and evaluation principles, as well as different solutions to their respective inherent concerns, there is a significant nexus between these two concepts that may contribute to their mutual success. Section 1.4 further elaborates on some studies discussing both concepts. Livability interventions signify the incremental phases that jointly expand the potential for longer-term progress towards sustainability. Creating livable communities lies within the horizon and responsibility of local agencies, planners, architects, and policymakers, who create environments where people’s needs and desires unfold and are satisfied.

1.3. International Livability Indices

Evaluation research on the performance of cities is an important source of valuable information for businesses, citizens, urban planners, and governments. Given that cities

are multi-faceted entities, they can be assessed from different perspectives, including productivity, quality of life, competitiveness, sustainability, innovation, and many other fields [31]. Most of the assessment studies use the indicator-based method and measure the performance of cities in a specific aspect by a three-step process: (1) defining a set of applicable indicators, (2) quantifying them, and (3) aggregating the indicators into a performance index [32]. The cities assessed include megacities, global cities, and local cities. The primary purpose of using indices is to benchmark existing conditions, measure progress, and increase accountability in integrated planning efforts to enrich community livability. Livability has evolved from being a social scientific index of the relative well-being of whole populations, i.e., a measure of the state of states, to being a measurable aspect of the subjective experience of individuals, i.e., an index of the state of persons [33]. Most researchers and policy makers consider quality of life to be self-explanatory and a reference for the standard of living or overall well-being of cities, with several international agencies having developed numerous approaches to understanding the standard of living of cities [34].

Among some notable global livability indices are the Economist Intelligence Unit (EIU) Livability Index [35] and the Mercer Quality of Life (QoL) Survey [36]. All these indices report on livability rankings of the living standards of various cities, and are examples of annual rankings [35,37,38]. In addition to the global rankings, there are those that are conducted at the regional level. Some include the Organization for Economic Co-operation and Development (OECD), which conducts the Better Life Index (BLI) to survey or measure livability based on eleven social–environmental–economic factors, and the American Association of Retired Persons (AARP), a public institute, which develops a livability index for American cities [39]. The information provided by these indices is used by policy makers to pinpoint and identify aspects of a city to make improvements where possible, to facilitate comparisons between cities, and to monitor the performance of the city over time. Figure 2 summarizes and compares aspects of QOL identified by the different indices.

Quality of Life Indices			
Mercer	EIU	Nation Ranking	OECD
Political and Social Environment	Material Wellbeing	Health Index	Housing
Economic Environment	Health	Education Index	Income
Socio-Cultural Environment	Political Stability and Security	Wealth Index	Jobs
Medical and Health Considerations	Family Life	Democracy Index	Community
Schools and Education	Community Life	Peace Index	Education
Natural Environment	Climate and Geography	Environment Index	Environment
Public Services and Transport	Job Security		Governance
Recreation	Political Freedom		Health
Consumer Goods	Gender Equality		Life Satisfaction
Housing			Safety
			Work-Life Balance
Each color represents similar or related indicators			

Figure 2. A Comparison Between QOL Indices.

International benchmarking studies can additionally support the growth of main industries in modern cities by developing and endorsing a city's image to attract highly skilled professionals, investment, tourists, and new citizens. Consequently, in the race to become the most livable city in the world, indices efficiently define what qualities are most critical in a city and/or country, as well as which qualities are critical to external stakeholders perceiving progress and assessing potential. City benchmarking improves the competitive positions of cities by providing information for strategic planning and urban policy making. Cities are key attraction hubs in terms of economic potential, community well-being, and hope for better standards of living. Individuals need to live in cities because they value the financial benefits, associated amenities, and levels of services. In this regard, the global ranking tools have touched serious issues linked to people's perceptions of livability and the extent to which cities meet their requirements and ambitions. As a result, city administrators seem to be increasingly interested in the concept of livability. The concept has become a state-of-the-art instrument for better definition of the future of cities.

1.4. Gap Analysis and Study Focus

Scholars and policymakers have extensively examined the concept of livability, producing a vast body of literature on the concept. as popular as the concept is in both research and practice, achieving it is a dynamic and complex process [22].

As noted earlier, the lack of a standard definition of livability hinders the comparison between cities/communities, where the use of a 'one size fits all' definition does not highlight differences between cities in terms of physical and social structures; for instance, the impact of a lack of infrastructure and the need to address the needs of vulnerable populations. Moreover, the majority of indices are empirical studies used to compare various geographic zones and do not explicitly state what the different indices measure or what their practical meaning might be for urban policy and planning and development [34]. Other research gaps include the need for simplification of each concept studied to a set of traits or characteristics and the fact that the tools used to assess livability do not necessarily consider residents' perceptions in the city. Moreover, while international benchmarking surveys can provide beneficial metrics and indicators for local decision-making, they take very little to no account of the perceptions of residents. As a result, the interpretations can be oversimplified and fail to show the reality of how a particular setting compares to others. This is a serious methodological weakness, as these perceptions are vital in evaluating the performance of any city [40]. Simultaneously, other 'social dimensions' of urban communities are increasingly included in the livability agenda, e.g., the diversity of populations and activities that add vitality to places and enhance personal and community experiences. According to Lloyd et al. [40], these qualities signify that a city is socially active, meaning that all citizens participate in social activities and events. However, this study argues that the social dimension of livability is multifaceted and difficult to define and operationalize, and hence, more difficult to measure in terms of livability policy outcomes.

Livable cities are places where people can delight in their desired or preferred quality of life. In the case of university campuses, they are unique due to their large size, population, and diversity of activities [41]. The unique characteristics of a university campus make it difficult to regard it as a city, neighborhood, or block [41]. University campuses typically include buildings with different use profiles, ranging from classrooms, offices, and laboratories to dormitories, restaurants, shopping, and sport facilities. Similar characteristics are observed in cities, except that they are based on population size and density. In both cities and universities, a high number of people use buildings on a daily basis, not only for research and educational purposes, but also for living and cultural participation activities. For these reasons, some studies have referred to universities as miniature cities [33]. The fusion of these elements contributes to fostering a livable, sustainable university campus that enhances the

quality of life of its residents. Livability indicators and assessments are crucial in promoting campus livability and sustainability. Given its large size and diversity of functions, the university campus shares several attributes with the urban space, such as circulation networks, built space, open space, and the arrangement and relations between these factors. Consequently, the indicators that are applied to urban space can also be applied to a university campus. Based on this, the livability indicators that apply to an urban form can be applied to the campus form. Livability affects every domain of a university, from the classrooms, to laboratories, to housing, to transportation, and to many other services, as well as the campus as a whole. Amongst the given definitions of a livable campus community, Habib [42] suggests that the protection and enhancement of the health and well-being of humans and ecosystems should be both local and global responsibilities exercised by actively engaging the knowledge of the university community in addressing ecological and social challenges.

Ensuring livability means identifying, mitigating, compensating, or offsetting the full costs of development proposals [43]. In recent years, the concepts of livability and sustainability with regard to university campuses are gaining more attention among practitioners and experts in the field [33,41–47]. Many of these studies focus on the design, planning, and construction stages [33,42,45–49], while others place emphasis on the environmental management practices [42,43] within and around the region. The campus planning practice has been the main topic for some of these academic studies [48–50]. Despite the various indicator frameworks, campus livability assessment in the middle east confronts many challenges, including lack of or limited access to data, and other difficulties in measuring indicators. In 1996, Dober stated that ‘Lacking an organized body of research or theory, campus planning is likely to be continued on a pragmatic basis’ [51]. On the other hand, presently universities are trying to apply sustainability initiatives on their campuses, yet a livability index to make campuses more livable is generally lacking. Studies in the region have already started the discussion and have highlighted the roles of campus operations and student training in promoting environmental sustainability [50,52]. Moreover, the idea of converting traditional campuses to smart campuses has been reported, finding that students believe that a sustainable environment is more important than the smart management of energy [53]. A study by Mushtaha et al. (2022) found that when university campuses consider sustainability, livability, safety, security, and comfort in all aspects of design, development, and operations, they can achieve the best designs [54].

The aim of this study is to improve and expand the state of knowledge by providing a theoretical framework to achieve the livability of university campuses. To measure university qualities is not an easy task, yet various surrogate variables have been used for this motive. Nevertheless, quantifying livable campus attributes has no precedent in the literature. This study is methodologically novel, compared to mainstream literature, in which it has conducted cross-sectional analyses in applying a combination of quantitative (survey) and qualitative (interviews with experts) research methods. This is the first study in the region that addresses the livability aspect of a university campus, and it compliments and expands on the previous studies.

2. Materials and Methods

2.1. Proposed Research Methodology

To enhance the precision of the results, a mixed methodology approach was adopted. The proposed index was developed based on several research methods, such as a literature review, the Delphi method of experts, and a case study. The following subsections explain the method in detail. The proposed methodology in this paper consists of sequential steps, as demonstrated in Figure 3.

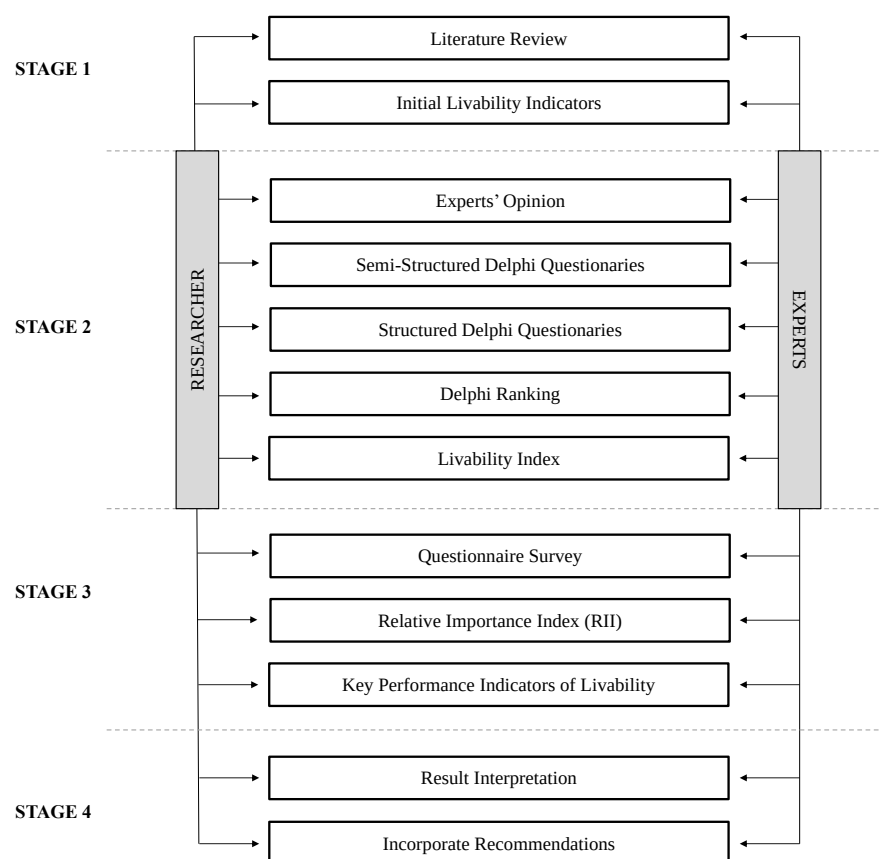


Figure 3. Proposed Research Methods for Assessing Livability Employed in This Study.

2.2. Indicators Inclusion Based on Literature Search

A preliminary list of indicators was proposed after reviewing the literature. These selected dimensions and their indicators were further validated by using semi-structured surveys with experts. The dimensions used in this study to assess the livability of university campuses were defined based on a literature review. The focus was on the identification of the various dimensions of livability that are suitable for the evaluation and assessment of urban areas. While there is a sensible consensus regarding the dimensions that should be used to assess livability, the indicators selected for each dimension vary from one study to another. There is no consistency on which aspects should be embraced to evaluate urban livability [4]. This study argues that the main reason for this is because the indicators usually reflect particular aspects specific to a particular context, which may not be applicable or relevant in other contexts. It is noteworthy that there is evidence that livability and quality of life are connected to perceptions, feelings, and subjective values [55]. The subjective approach of measuring livability involves making subjective assessments of individuals' satisfaction with their living conditions based on questionnaires survey. Most of the livability studies used the subjective evaluation to build an index system (e.g., [56–58]). Consequently, this study uses the subjective approach to build the evaluation framework for livability at a university campus level.

For practicality purposes, 3 criteria were used to assess the indicators in this research and determine their eligibility for inclusion:

- The indicator must be important to livability and/or the social factors of health and wellbeing in urban areas.
- The indicator must be relevant to university campus assessment.
- Studies that have identified livability indicators must look at different parts of the world, not only one country.

After identification, indicators were classified into three categories based on these inclusion criteria:

- Favorable and meets all or most of the criteria.
- Beneficial but needs further development to meet the inclusion criteria.
- Not beneficial because it either fails to meet the inclusion criteria or is unnecessary because of similar, more promising measures.

The final selection of indicators was divided into 6 dimensions (Environmental Quality, Social and Cultural Elements, Accessibility, Safety, Amenities and Services, and Housing). There was a degree of subjectivity in the creation and application of these criteria. However, previous work suggests this approach is appropriate because livability is a subjective and relatively new concept [59]. Further, livability was considered from many perspectives, as multidisciplinary experts contributed their expertise in urban planning and design, engineering, and sustainability. The composed criteria were further filtered and categorized to fit the context of the study, the UAE, through the application of the Delphi method.

2.3. Expert Selection of the Delphi Method for Consensus

2.3.1. Delphi Method

The Delphi method is typically used to survey a panel of experts in order to reduce diversity of opinions within a small group and reach consensus [60]. The Delphi method [61] was chosen as being the most appropriate approach to meeting the research purpose of this paper. The Delphi method was used in this study to repeatedly acquire expert opinions until there was consensus on the selection of projects, prediction of problems, and resolution of problems, as outlined in Figure 4. Subjective measures were used to directly explore the experts' opinions. As stated by Maleki et al. [8], surveys often attempt to identify 'what is', while the Delphi method attempts to address 'what could/should be'. Additionally, the Delphi method was used to modify the indicators through the following criteria:

- Relevance and Impact: The indicators are related to one or more matters that are cared about and that have significant policy impacts.
- Simplicity and Clarity: The measures are exciting and comprehensible to the general public and to policy makers.
- Comparability: General enough so that the community, in this case university campuses, can be compared to one another.

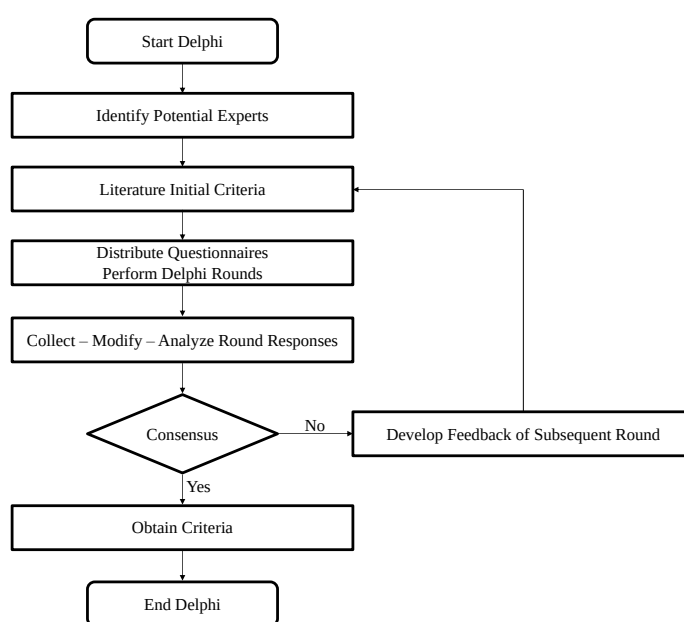


Figure 4. Delphi Method.

2.3.2. Expert Selection for Delphi and Application

To ensure a valid study, the Delphi method requires a focus on the selection of individuals, who should have backgrounds and experience related to the targeted topic in order to be qualified to participate. For this study, the selection of experts was based on purposive sampling, where participants were deliberately selected based on their knowledge and experience with the research topic, as well as their willingness and availability to participate in the research. The experts were selected based on their publication record in the areas of sustainability and livability, experience in both the academic and industrial fields, time of expertise in the sustainability/livability fields, and knowledge on the local, regional, and global scales. The number of experts included in a Delphi panel can vary greatly. For example, according to Thangaratinam and Redman [61], studies may include as few as 4 participants. Consensus is likely to be reached within a faster timeframe with smaller panel sizes. In the current study, the size of the Delphi panel was determined by some pragmatic factors, including but not limited to the question to be answered, timeframe of study, delivery method, and access to experts and resources. Four experts were selected in this study, and their expertise included sustainable community planning, land use, environmental planning, urban economic development, urban traffic operations, control, public health–transport systems interactions, spatial planning, urban planning and design, land use and development regulations, sustainability planning and design, sustainable development, construction engineering, and project management. All selected experts had over 15 years of experience within the American University of Sharjah Campus in the United Arab Emirates, but also at an international level, including United States, Canada, United Kingdom, Australia, Germany, Nigeria, Sudan, and Colombia, amongst others. All experts provided useful inputs and reviewed their initial/previous decisions to achieve consensus.

The Delphi survey was conducted in five steps: (1) identifying the different aspects of information required and the group of experts; (2) distributing the first survey to the experts; (3) collecting the first survey, compiling it, and analyzing the experts' responses; (4) distributing the second survey to the experts; and (5) collecting the second survey, compiling it, and analyzing the experts' responses. A total of three rounds of Delphi Method were required to complete the evaluation of the prospective indicators using a 5-point Likert Scale. After each round, the survey was modified based on the experts' advice, and it presented the preceding results anonymously to allow the experts to reevaluate the responses without peer pressure. Moreover, experts assisted in determining whether dimensions and/or indicators should be added, rejected, or modified based on the scores and recommendations. The research applied the standard coefficient of variation (CV) to assess the consensus among experts to evaluate the indicators as depicted in Equation (1). When the $CV \leq 0.3$, it indicates that the experts agree with one another [62–64].

$$\text{Coefficient of Variation (CV)} = \frac{s}{\bar{X}} \quad (1)$$

2.4. Study Area: American University of Sharjah (AUS)

The research used the American University of Sharjah (AUS), which is an independent nonprofit educational institute located in Sharjah, United Arab Emirates (UAE), as a case study. AUS is known for its distinguished teaching, learning, research, scholarships, educating, and mentoring of future leaders [65]. AUS is ranked first worldwide for diversity, with an employability rate of 88%. It is among the top 50 universities in the world under the age of 50 years, according to QS world ranking, and has been in the top 10 in the Arab world for 5 years. AUS includes a wide range of Bachelor's, Master's, and Ph.D. programs for several colleges: the College of Engineering, the College of Architecture, Art and Design, the School of Business, and the College of Arts and Sciences. Table 1 presents the AUS statistics for 2022. In line with sustainable development, and to continue the vision and legacy of the founding father of the UAE, the late Sheikh Zayed Bin Sultan Al Nahyan, AUS Sustainability was established in 2009 to incorporate sustainability at all levels of the campus by leveraging collective students, staff, faculty, alumni, and partners [65].

AUS Sustainability integrates sustainable thinking and action into campus operations, teaching, and research [53,66] by working with all campus stakeholders, drawing on diverse academic and professional backgrounds.

Table 1. AUS Statistics for 2022.

Details	Total Count
Undergraduate Students	4691
Masters' Degree Students	425
PhD. Students	68
Achievement Academy	105
Total Student Enrollment—Headcount	5289
Students Residing in Dormitories	1125
Faculty, Staff, and Dependents Residing on Campus	1187

The campus covers over one million square meters of land. Its location and coverage are shown in Figure 5A,B. Thus, it can be considered as a microcosm of a city. The importance of livability cannot be overstated. Assessing the livability of the AUS campus will ensure that the university is taking steps towards sustainability and enhancing the quality of life (QoL) of its residents. The university can be subdivided into three major areas: the student residential accommodation known as dormitories, the academic area, and the faculty area, which houses the staff.

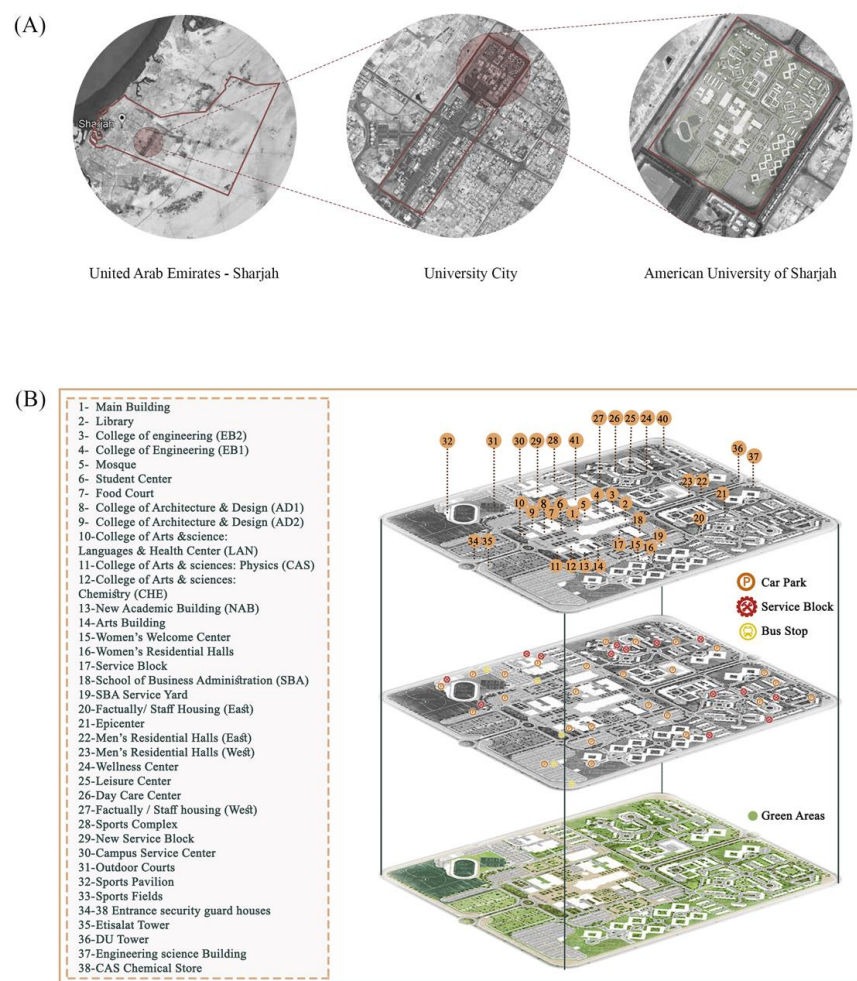


Figure 5. American University of Sharjah (AUS). (A) Location, and (B) campus axonometric diagram.

2.5. Data Collection

2.5.1. Questionnaire for AUS Residents

A cross-sectional Survey-Monkey-designed questionnaire was adopted. The questionnaire was distributed among the stakeholders, which included students, staff, and faculty of AUS for a duration of 13 days. The sample comprised 267 randomly selected participants. The questionnaire was used to measure the current livability of AUS from the residents' point of view, and it included demographic information and a set of questions to acquire information about the residents' satisfaction with the livability conditions at AUS. The respondents were asked to rate their satisfaction using a five-point Likert scale. This scale comprises a range of responses used to define levels of agreement. Respondents could also provide any additional comments they had through one open-ended question. The responses on a five-point Likert scale ranged from 5 = Strongly Agree to 1 = Strongly Disagree. Finally, Relative Importance Index (RII) was used to identify the most important criteria based on the participants' perceptions.

2.5.2. Validity, Reliability, and Assumptions

Face validity was used to ensure that the survey was appropriate for the purpose, comprehension, and consistency of this study. In addition to testing face validity with a sample of leading experts in the field, a group of unbiased people were also used in order to control the potential effect of literacy on reading comprehension. The questions included in the survey were clear and included explanations for some of the concepts to ensure the understanding of the readers. It is worth noting that, before deploying the questionnaires to the AUS community, the questionnaire was revised by experts for suggestions that aided in improving the instruments, and a pilot group answered the questionnaire for testing purposes and to ensure that no bias existed. Respondents were reassured of the anonymity of the responses. Content validity was validated by ensuring that the content of the surveys conducted in this study covered all relevant aspects of the subject it aimed to measure; the subject being the campus livability index [64]. Cronbach's alpha α was used to determine the internal consistency, where the result was a number between 0 and 1. Cronbach's alpha measures the extent to which item responses, namely the answers to survey questions, correlate with one another. In other words, Cronbach's alpha estimates the proportion of variance that is consistent in a set of survey answers. By convention, statisticians claim that an alpha of (0.65–0.80) is regarded as 'adequate' for a scale used in human scope research [67,68].

The following assumptions were made in this study: the experts were well-informed about livability and sustainability; expert responses were honest and reliable based on their knowledge and experience; and all responses from survey takers (AUS stakeholders) were candid.

2.6. Index Calculations Relative Importance Index (RII) and Livability Index

Relative Importance Index (RII) was performed to prioritize the significant factors, followed by a reliability analysis to check the consistency of the data. The sample size obtained in this study was adequate to accomplish its goal, as the RII requires a minimum of 100 respondents to obtain valid results [69]. To compare the relative importance of each indicator to the maximum importance it could achieve, the ranking survey's data were analyzed using the RII, which was calculated using Equation (2). Here, (w) is the total weight given by respondents for each indicator, ranging from 1 to 5, multiplied by the number of respondents (n_1, n_2, n_3, n_4, n_5). To illustrate, n_1 represents the number of respondents for 'Strongly Disagree'; n_2 represents the number of respondents for 'Disagree'; n_3 represents the number of respondents for 'Neutral'; n_4 represents the number of respondents for 'Agree'; and n_5 represents the number of respondents for 'Strongly Agree'. The highest

weight is indicated by the A (5 in this case), and N indicates the number of respondents (270 responses in this case).

$$\text{Relative Importance Index (RII)} = \frac{\sum w}{AN} = \frac{5n_5 + 4n_4 + 3n_3 + 2n_2 + 1n_1}{5N}; (0 \leq \text{RII} \leq 1) \quad (2)$$

For the purpose of this paper, the overall livability of a city/neighborhood/community was used to assess the livability of a campus. The authors considered that although universities are not necessarily cities or neighborhoods, their sense of community makes the approach applicable to this study. The overall livability is rated on a scale of 0 to 100 by the livability index score. Based on the average of all seven category scores, which likewise range from 0 to 100, the overall livability score is calculated. The scores are determined by comparing the communities to one another; as a result, the average community receives a score of 50, while communities that are above average and below average receive higher and lower scores, respectively. The objective is to assist individuals and community leaders in learning about their neighborhoods and determining if they offer the amenities that residents desire and require. Stakeholders can then utilize the Index to act to address gaps and improve livability.

$$m = \frac{\text{Sum of terms}}{\text{Number of terms}} > \frac{1}{m} \quad (3)$$

3. Results

3.1. Indicator Determination Based on Literature

Through various studies and a comprehensive literature review, the determining indicators affecting people's satisfaction with urban livability are summarized in Table 1. A total of 37 indicators are defined under seven main dimensions, and were used to create the university campus livability index. There was general agreement that indicator selection was often subjective, and the choice of an indicator depended on factors such as whether it was cost-effective, easy to understand, scientifically reliable, and internationally comparable. Commonly used indicators, such as crime rate and safety, accessibility/transportation, health, natural environment, and service facilities were considered. Moreover, indicators that reflected urban features of AUS were also incorporated, such as places of worship, banking services, and postal services. Table 2 illustrates the final list of indicators after the validation process by experts was performed, which is described in the following sections. It indicates that this is the first study that comprehensively and simultaneously evaluates livability perceptions through indicators for achieving a livable university campus, which signifies the importance of this paper. Although the present study focuses on AUS as a case study, the methods utilized can be replicated for other institutions globally.

Table 2. Dimension Index of a Livable Campus based on Literature.

Dimension Index	Previous Studies																			
	[1]	[2]	[3]	[70]	[71]	[72]	[73]	[30]	[74]	[75]	[76]	[77]	[78]	[79]	[80]	[4]	[81]	[82]	[83]	[84]
Environmental Quality																				
Tree Shading	X	X	X			X	X	X				X				X	X	X		X
Landscaping	X	X	X			X	X	X		X		X				X	X	X		X
Outdoor Lighting		X	X							X				X		X	X	X		
Public Seating Amenities	X		X		X			X	X				X	X		X				
Street Maintenance/Cleanliness.	X		X	X	X				X	X										
Social and Cultural Elements													X							
Leisurely Gathering	X	X	X	X		X		X	X			X	X	X	X		X	X		
Cultural Events		X			X	X		X	X			X								X
Social Gatherings			X	X		X					X		X	X	X			X		
Socio-cultural Competitions for Kids					X	X									X		X			X
Green areas, parks, playgrounds for kids	X		X		X	X						X				X				X

Table 2. Cont.

Dimension Index	Previous Studies																			
	[1]	[2]	[3]	[70]	[71]	[72]	[73]	[30]	[74]	[75]	[76]	[77]	[78]	[79]	[80]	[4]	[81]	[82]	[83]	[84]
Accessibility																				
Sidewalk Physical Condition	X	X	X		X	X				X	X		X			X	X		X	
Accessibility for Special Needs			X				X		X	X	X									
Parking Spaces	X	X	X		X		X		X	X						X	X			
Wayfinding Signage				X						X										X
Campus Connectivity to the Surroundings				X		X				X		X	X	X		X	X	X		
Public Transport around Campus	X	X	X		X	X	X		X	X			X			X			X	
Safety																				
Crime Rate	X		X		X		X	X	X	X	X	X	X	X	X	X	X		X	
Visibility of Security Officers					X			X	X	X	X									
CCTV Cameras					X		X		X	X		X			X					
Threats from Untamed Pets																				
Gender Harassment Outdoors			X		X						X						X			
Amenities and Services																				
Healthcare Services	X	X		X	X	X	X	X	X		X		X		X		X			X
Facilities for Special needs		X			X		X				X		X							
Sports and Recreation Facilities	X	X		X	X		X		X				X	X			X			X
Emergency Services on Campus																				
Quality of Technology												X		X						
Housing																				
Housing Maintenance Services	X	X	X		X	X	X						X	X	X		X			
Accessibility to Residential Areas (Walking/Cycling)	X		X		X	X		X			X	X	X				X		X	X
Sense of Privacy in the Dwelling Units	X				X								X		X				X	
Quietness of Residential Complexes	X					X								X					X	
Neighborliness/Friendliness of Residents			X												X					
'Other' Services																				
Places of Worship													X		X					
Banking Services	X			X	X												X			
Postal Services	X			X	X												X			
Veterinary Services	X			X	X												X			
Public Transport to off Campus Places	X	X	X		X	X		X					X				X		X	X
Services and Utilities Provided for Building Users	X	X			X								X				X		X	X

X: The assessment was found in the study.

3.2. Delphi Analysis and Experts' Results

The Delphi surveys from Rounds 1, 2, and 3 were completed by four specialists in sustainability and livability (with 100% response rates), and the results were used to calculate the livability indices for the AUS campus. Participants elucidated, in written comments, that several criteria could affect AUS campus planning, and advised a set of more essential criteria. Some criteria were eliminated because there were similar conditions throughout the study area. Appendix A presents the Delphi process undertaken to arrive at the final list of indicators. A total of 37 indicators were defined under seven criteria in the last round. The authors added 'Other Services' and its indicators to support adequate urban services and infrastructure that fulfilled the residents needs and expectations. Moreover, it should be noted that some of the initial indicators were linguistically modified by the experts. The final list of livability indicators obtained by the experts through Delphi Method were analyzed by computing the average (mean) and the standard deviation (the lower value is the better) using the Likert 1–5 scale (Least Important to Most Important) scores given by the experts. Each indicator met the required levels of Coefficient of Variation (CV) (less than 0.3) and is shown in Table 3, where the indicators selected by experts are rearranged in descending order from the most important indicator, as per the experts responses, to the least important one. This reveals that, using the evaluative standards, the investigation into the experts' responses reached a consensus. The highest coefficient of variation value was 0.28 for 'Healthcare services', 'Sports and recreation facilities', and 'Postal services. On the other hand, the lowest CV value was zero for 'Crime rate', 'Emergency services on campus', 'Quality of technology', and 'Housing maintenance

Services', which means all the experts agreed that these were the most important indicators in the study.

Table 3. Delphi Analysis Results.

No.	Indicators Defined by Experts	Mean ($\bar{X}$)	Standard Deviation (s)	Coefficient of Variation (CV)
1	Wayfinding signage	4.75	0.5	0.10
2	Neighborliness/friendliness of residents	4.5	0.57	0.12
3	Socio-cultural competitions for kids (beauty contests, educational debates, etc.)	3.75	0.5	0.13
4	Places for informal leisurely gathering on campus (cafes, restaurants, etc.)	3.5	0.57	0.16
5	Veterinary services	3.5	0.57	0.16
6	Organized cultural events for different countries (art, fashion, food, etc.)	3	0.57	0.27
7	CCTV cameras	2.75	0.5	0.18
8	Threats from tamed and untamed pets	2.75	0.5	0.18
9	Various housing choices	2.75	0.5	0.18
10	Amount of public seating amenities	2.5	0.57	0.23
11	Amount of landscaping	2.5	0.57	0.23
12	Public transport around campus	2.5	0.57	0.23
13	Quietness of residential complexes	2.5	0.57	0.23
14	Public transport to off campus places	2.5	0.57	0.23
15	Organized social gatherings for special occasions	2.25	0.5	0.22
16	Campus connectivity to the surrounding areas	2.25	0.5	0.22
17	Visibility of security officers	2.25	0.5	0.22
18	Facilities for special needs	2.25	0.5	0.22
19	Sense of privacy in the dwelling units	2.25	0.5	0.22
20	Places of worship (mosques, etc.)	2.25	0.5	0.22
21	Banking services	2.5	0.57	0.23
22	Amount of outdoor lighting	1.75	0.5	0.27
23	Sidewalk physical condition (e.g., firm surface, slip-resistant, and free from cracks, etc.)	1.75	0.5	0.28
24	Accessibility for the disabled	1.75	0.5	0.23
25	Healthcare services	1.75	0.5	0.28
26	Sports and recreation facilities	1.75	0.5	0.28
27	Postal services	1.75	0.5	0.28
28	Services and utilities provided for building users	1.75	0.5	0.23
29	Parking spaces	1.5	0.57	0.13
30	Accessibility to residential areas (by foot)	1.25	0.5	0.18
31	Amount of tree shading	1	0	0
32	Street maintenance and cleanliness.	1	0	0
33	Crime rate	1	0	0
34	Gender harassment outdoors	1	0	0
35	Emergency services on campus	1	0	0
36	Quality of technology (Wi-Fi, phone, cable, etc.)	1	0	0
37	Housing maintenance Services	1	0	0

Furthermore, all the standard deviation values for all the livability indicators are close to 0.5, which indicates that most of the rating values are close to the mean ($\bar{X}$) with a value of ± 0.5 (on average). Thus, after analyzing the Delphi method results, the livability indicators were utilized to develop the livability index for AUS campus.

3.3. Livability Perceptions

Questionnaire Results and Reliability

A sample size of 267 students, faculty and staff from the AUS was randomly selected. The demographic information for the respondents shows that almost 25% of the respondents were males, while females contributed to most of the responses, totaling almost 74%. In addition, the age of most of the respondents ranged from 25 to 40 years (55.63%). For

marital status, it was observed that most of the respondents were married (51.68) and their family size ranged between 2–4 people living in the same accommodation. The majority of the respondents were students (51.33%). As for education, the majority of the respondents reported that they had a master's degree (49.66%) while the minority had a high school diploma (2.68%). In terms of the years living on campus, more than 41% of the respondents had lived in the campus for more than five years. Finally, 96% of the respondents were non-Emiratis, while only 4% were Emiratis.

To assess reliability, Cronbach's alpha coefficients were used. First, a reliability analysis was carried out to assess the internal consistency of the questionnaire as a whole. Cronbach's alpha showed that the questionnaire reached excellent reliability ($\alpha = 0.92$). Then, the calculations for Cronbach's alpha coefficients were derived for seven criteria; namely, Environmental Quality, Social and Cultural Elements, Accessibility, Safety, Amenities and Services, Housing, 'Other' Services based on the information supplied by the 267 respondents. Table 4 presents the computation results. All indicators across the seven criteria had Cronbach's alpha coefficients greater than 0.65. As a result, the information from the questionnaire is regarded as reliable.

Table 4. Cronbach's Alpha Results.

Criteria	Cronbach's Alpha (α)
Environmental Quality	0.820575
Social and Cultural Elements	0.846369
Accessibility	0.784669
Safety	0.656212
Amenities and Services	0.714571
Housing	0.94222
'Other' Services	0.845861

3.4. Relative Importance Index (RII) Analysis

In Table 5, the average scores obtained from the questionnaire are combined to make up the RII values for the seven identified criteria and 37 associated indicators of the Livable Campus. The RII values used were as suggested by Rajgor et al. [69]: High (H) ($0.8 \leq \text{RII} \leq 1$), High–Medium (H–M) ($0.6 \leq \text{RII} \leq 0.8$), Medium (M) ($0.4 \leq \text{RII} \leq 0.6$), Medium–Low (M–L) ($0.2 \leq \text{RII} \leq 0.4$), and Low (L) ($0 \leq \text{RII} \leq 0.2$). It can be noted from Table 4 that the highly important indicators ($0.8 \leq \text{RII}$) from the stakeholders' standpoint include: amount of landscaping, street maintenance and cleanliness, accessibility for the disabled, crime rate, gender harassment outdoors, various housing choices, accessibility to residential areas, and services and utilities provided for building users. Additionally, the rest of the indicators were also above the 0.65 RII value and within the H–M range. This implies and validates that all the seven designated criteria and the associated indicators are important of a Livable Campus. This indicates that the results were quite in line with the chosen indicators for a Livable Campus.

Table 5. Results obtained for the Relative Importance Index (RII).

No.	Indicator	RII	Level of Importance
1	Crime rate	0.96	H
2	Visibility of security officers	0.89	H
3	Street maintenance and cleanliness	0.89	H
4	Amount of landscaping	0.86	H
5	Housing maintenance Services	0.86	H
6	Services and utilities provided for building users	0.85	H
7	Gender harassment outdoors	0.84	H
8	Accessibility for the special needs	0.83	H
9	Sports and recreation facilities	0.83	H
10	Accessibility to residential areas	0.83	H

Table 5. Cont.

No.	Indicator	RII	Level of Importance
11	CCTV cameras	0.81	H
12	Quality of technology	0.81	H
13	Amount of outdoor lighting	0.80	H
14	Postal services	0.80	H
15	Healthcare services	0.79	H-M
16	Veterinary services	0.79	H-M
17	Parking spaces	0.78	H-M
18	Emergency services on campus	0.78	H-M
19	Campus connectivity to the surrounding areas	0.77	H-M
20	Amount of tree shading	0.76	H-M
21	Neighborliness/friendliness of residents	0.76	H-M
22	Banking services	0.76	H-M
23	Sense of privacy in the dwelling units	0.75	H-M
24	Wayfinding signage	0.75	H-M
25	Sidewalk physical condition	0.75	H-M
26	Quietness of residential complexes	0.75	H-M
27	Places of worship	0.74	H-M
28	Various housing Choices	0.73	H-M
29	Threats from tamed and untamed pets	0.71	H-M
30	Organized cultural events for different countries	0.70	H-M
31	Places for informal leisurely gathering on campus	0.69	H-M
32	Organized social gatherings for special occasions	0.69	H-M
33	Public transport around campus	0.69	H-M
34	Facilities for special needs	0.68	H-M
35	Amount of public seating amenities	0.66	H-M
36	Socio-cultural competitions for kids	0.63	H-M
37	Public transport to off campus places	0.61	H-M

3.5. Indicators and Their Importance

Indicators were grouped based on their categories. Livability is a multi-dimensional concept. Although the dimensions may vary between different studies, the three dimensions of livability are often explicitly or implicitly covered in many previous studies, including the social, economic, and environmental/physical dimensions. For each defined dimension, indicators should be included as the impact factors that may affect the evaluation from many perspectives. Therefore, based on the comprehensive literature review and Delphi expert panel, the final list of indicators was grouped into seven categories/dimensions that fit university campuses, namely environmental quality, social and cultural elements, accessibility, amenities and services, safety, housing, and ‘other’ services. Figure 6 shows the final proposed livability campus framework based on the participation of experts by validating the indicators deduced from the literature as they deemed it necessary and essential to the livability of university campuses.

The results of the questionnaire were further explored to understand the respondents’ preferences. The average scores for the Environmental Quality criterion, including five indicators, are shown in Figure 7A. The results showed that respondents were mostly satisfied with street maintenance and cleanliness at AUS (45.75%), which was followed by the amount of landscaping (41.45%), amount of outdoor lighting (28.76%), amount of tree shading (16.99%), and amount of public seating amenities (10.46%). Figure 7B presents the average scores for the accessibility criterion, including five indicators. The results indicated that respondents were somewhat satisfied with this criterion, as the majority of the participants answered ‘Agree’ for all the indicators. However, regarding the ‘Public transport around campus’ indicator, the number of participants was almost equal for ‘Agree’ and ‘Neutral’. This indicates that ‘Public transport around campus’ indicator was the weakest in this criterion. According to the results, the only motivating factors for commuting by foot were safety and continuity along pedestrian paths. Respondents noted

that lighting, the quality of the pavement, accessibility by users with special needs, shading elements, pedestrian crossing and signage, width of pedestrian pathways, and quality of crosswalks on the pedestrian pathways directly affected walking efficiency.

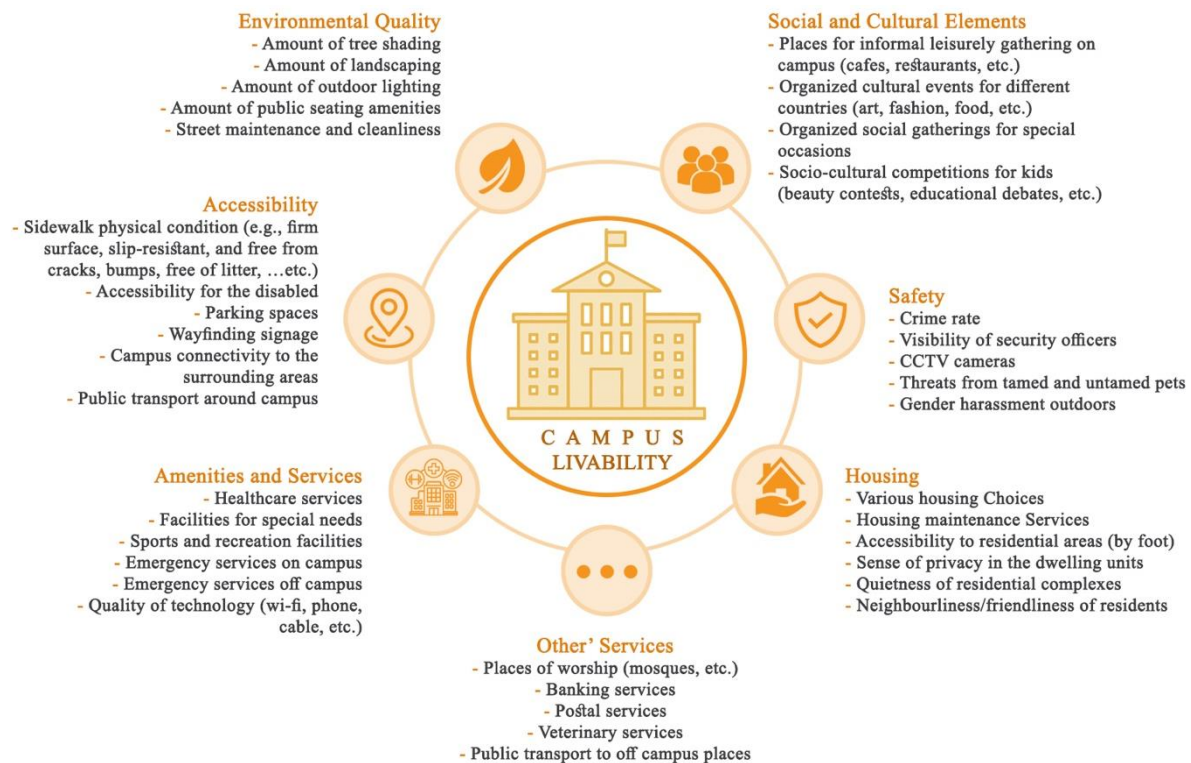


Figure 6. Proposed Livability Campus Framework.

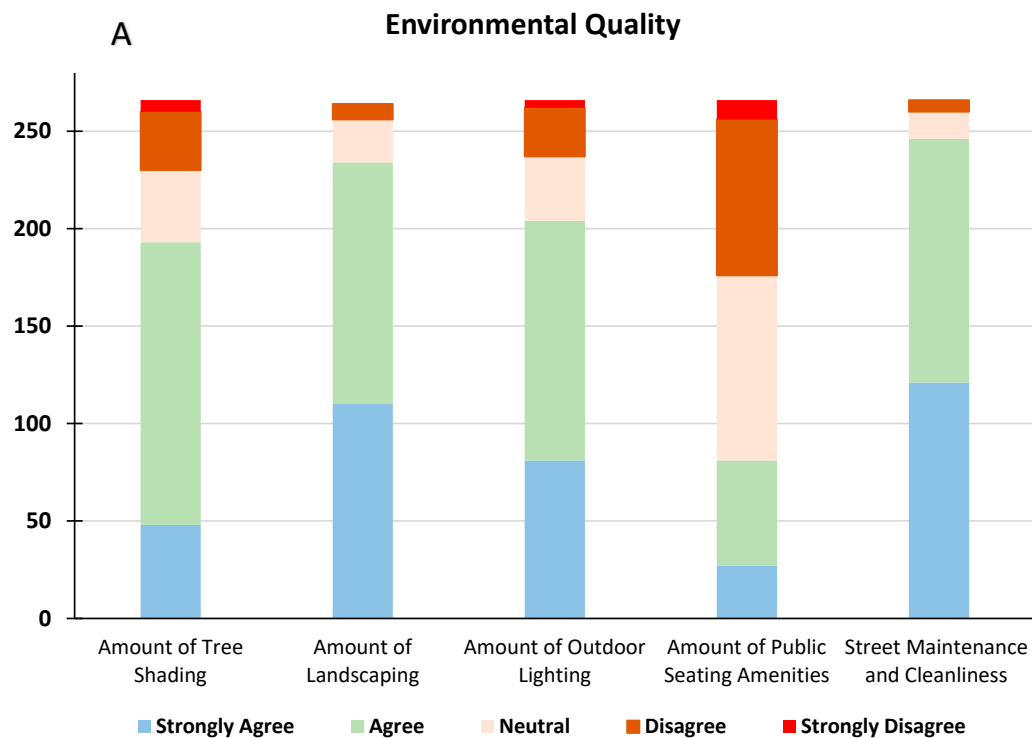


Figure 7. Cont.

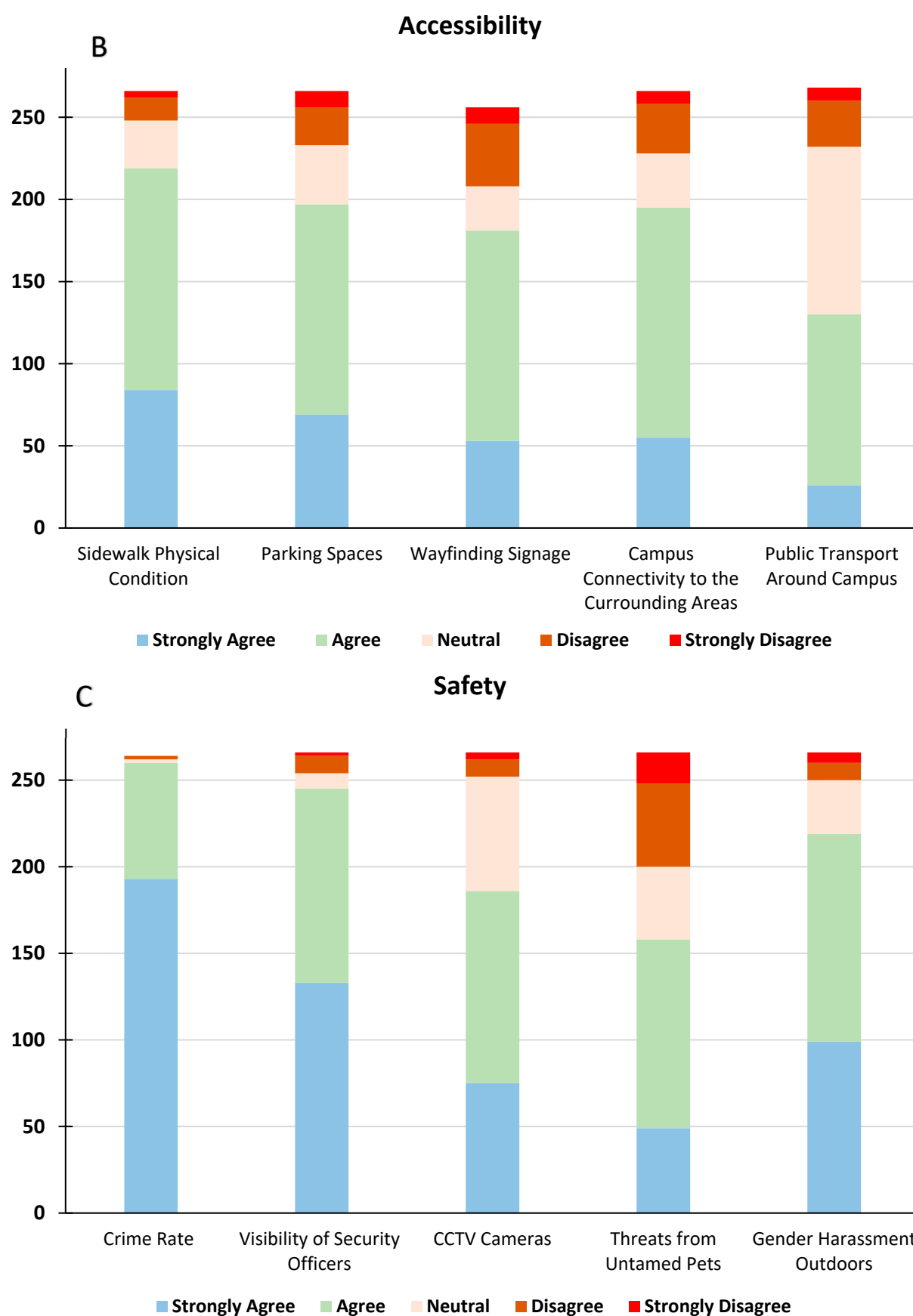


Figure 7. Responses to each indicator based on their preference to the AUS Campus. (A) Environmental Quality; (B) Accessibility; (C) Safety.

Safety is a significant factor for the livability of any community. On the AUS campus, the average scores for the safety criterion, including five indicators, are shown in Figure 7C. The majority of respondents felt quite safe at AUS campus and felt that security officers on campus were providing effective services. However, a total of 35 participants (23%) answered 'Disagree' and 'Strongly Disagree' for 'Threats from untamed pets' and 'Gender harassment outdoors' indicators. Therefore, these two indicators must be addressed by AUS authorities in order to increase the livability status of the campus. In order to maintain the health status of individuals and to protect their human rights, safety is essential. The importance of making campuses safe havens for students, faculty, and staff should, therefore, be emphasized. People's wellbeing is affected by their housing, which contributes to their physical and psychological health, their safety and security, and their need for love and belonging. At AUS, the majority of the faculty and a large percentage of students live on campus. Figure 8 presents the average scores for the housing criterion including seven indicators. Figure 8 indicates that almost all participants considered the housing area accessible (58%). Moreover, the results illustrate that the highest ranking was 4 'Agree' for all the indicators in this criterion. In terms of neighborhood friendliness, the respondents' answers indicated that when the neighbors were friendly and the houses were sustainable, they definitely felt that their houses were livable. On the other hand, 33 participants ranked the 'Quietness of residential complexes' 3 'Neutral', and 28 participants ranked the 'Sense of privacy in the dwelling units' 3 'Neutral'. Therefore, AUS residents were not really satisfied with these indicators, and this must be addressed by AUS authorities in order to increase the livability status of the campus. It is worth noting that, in terms of livability, respondents believed that socio-psychological needs, as well as inbuilt housing facilities and public amenities, had a significant impact.

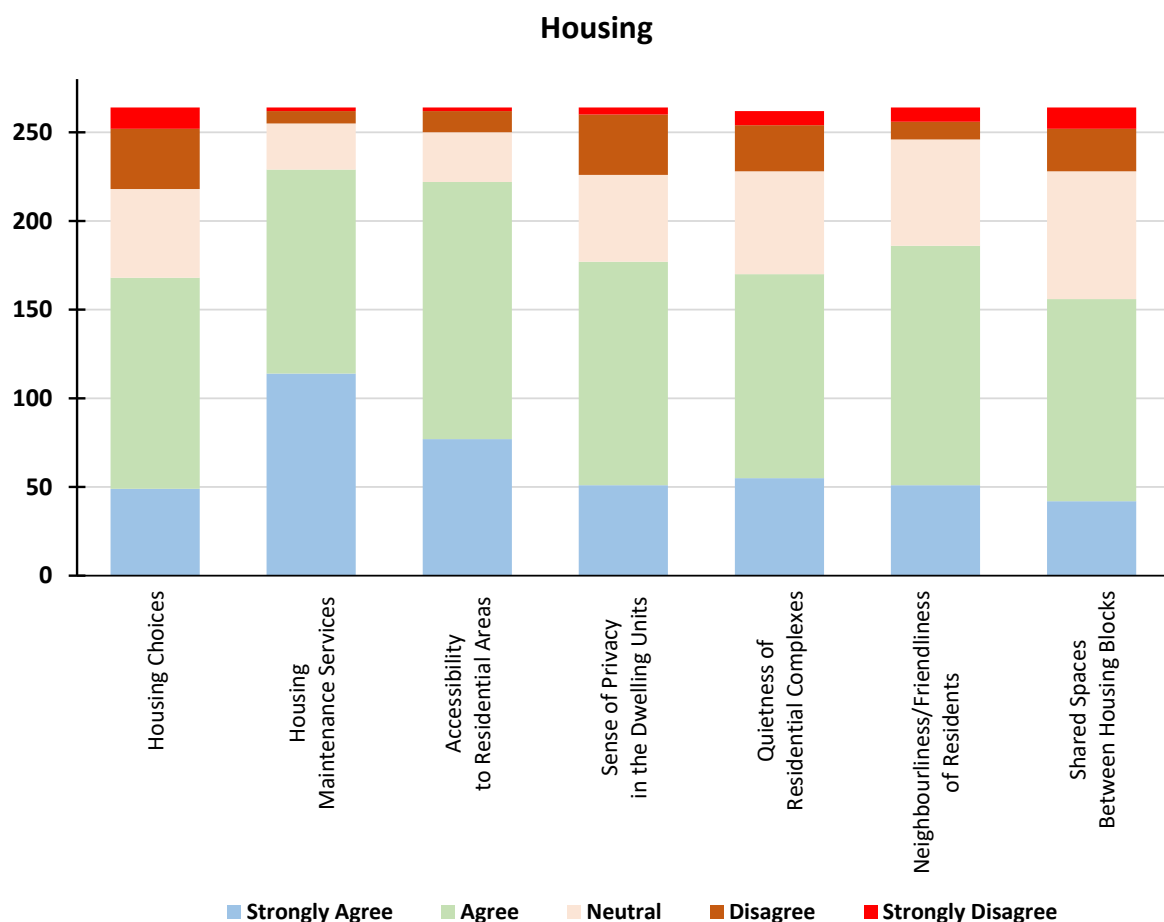


Figure 8. Responses to the housing indicator based on their preference to the AUS Campus.

One of the most important aspects of a livable community is the provision of amenities and services. A main factor in assessing this is convenience, which includes improved access to amenities and services. As shown in Figure 9A, the average scores for the amenities and services criterion, including five indicators, indicated that all the respondents answered 'Agree' for all the indicators in this criterion. For 'Facilities for special needs' indicator, 64 participants (43%) answered 'Neutral'. This means that AUS achieves the goals of livability under this criterion, such as ensuring effective service delivery of necessary services and infrastructure. However, the majority of the respondents were not really satisfied with the 'Services for Special Need' indicator. Another criterion assessed was intended to reveal how the AUS campus inspires the formation of better-quality levels of social and community interactions, i.e., social life. Figure 9B presents the average scores for the social and cultural elements criteria, including five indicators. The results indicated that respondents were not really satisfied with this criterion, as the majority of the answers lay between 'Agree' and 'Neutral'. Participants highlighted that they enjoyed parks and walking, and that the public realm of the AUS campus should be further developed to enhance social activities. Therefore, there was a need to enhance these measures at AUS to increase its livability.

The final indicator, presented in Figure 9C, is the average scores for the 'other' services criterion, including seven indicators. The results indicated that the highest ranking was 4 'Agree' for all the indicators in this criterion, except for 'Public transport to off campus places' indicator, where 70 respondents answered 'Neutral'. In general, universities should implement green transportation systems that provide bus services and pedestrian walkways on campus, encourage carpooling and bicycling among the community, and charge for on-campus parking.

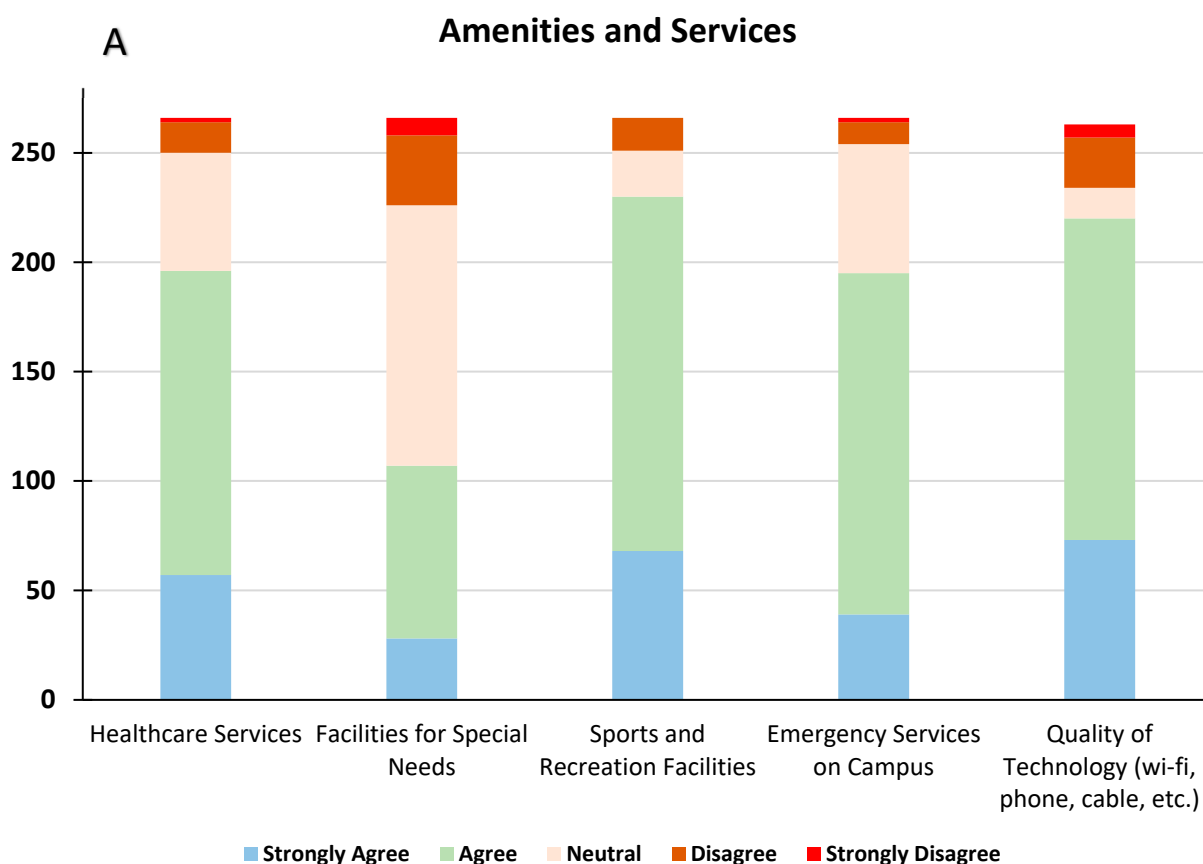


Figure 9. Cont.

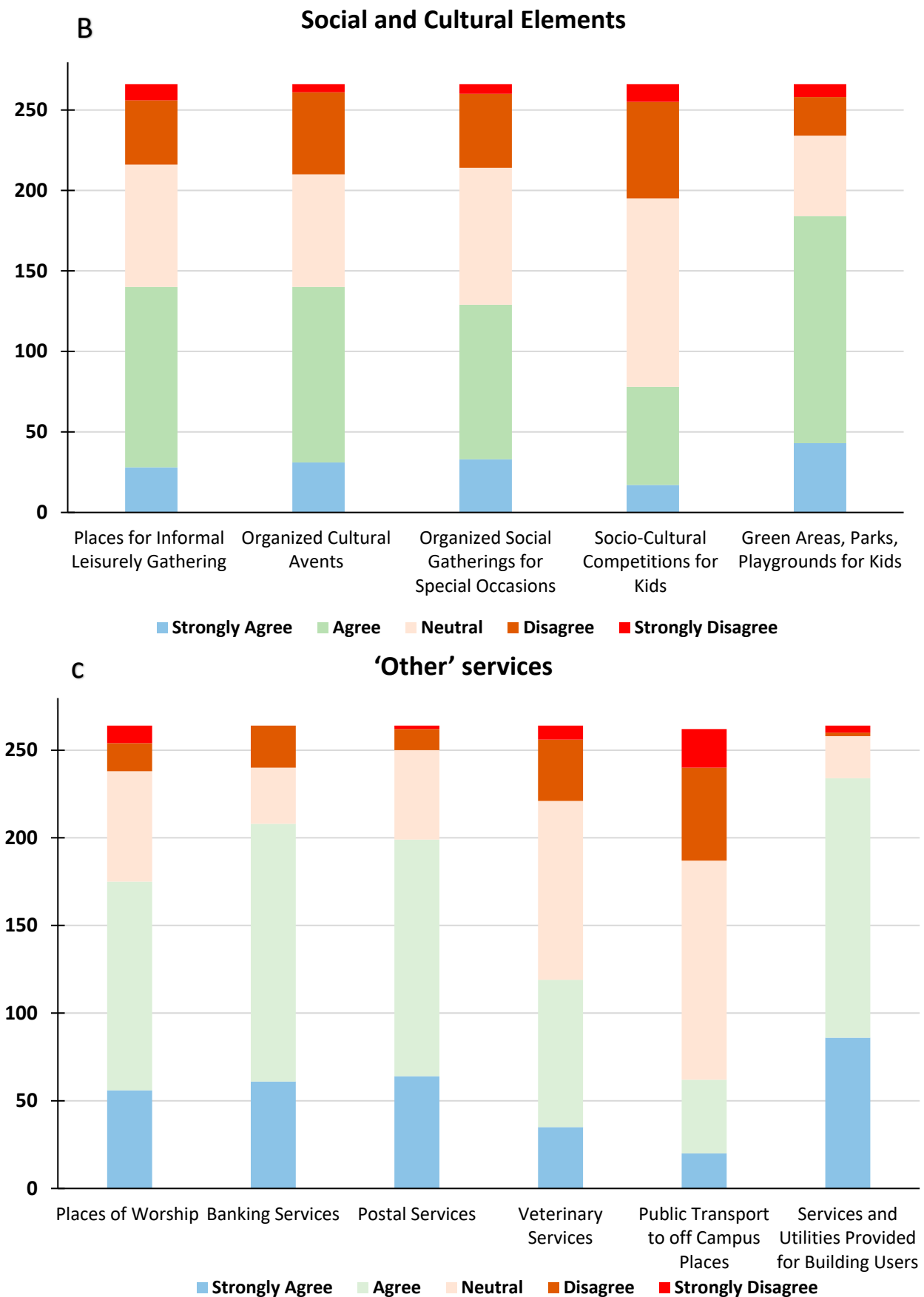


Figure 9. Responses to each indicator based on their preference to the AUS Campus. (A) Amenities and Services; (B) Social and Cultural Elements; (C) Other Services.

The presence of opportunities, facilities, and amenities in space, apart from their sheer distance, becomes applicable if the following conditions are in place: (1) the space is accessible by foot or by bicycle; (2) the space has enjoyable connective pedestrian paths that are spatially integrated with the surroundings; (3) the space is provided with urban activities, and (4) the space does not prioritize car traffic, either by design or in response to the main social interactions occurring there.

3.6. Overall Assessment of Livability at the AUS Campus and Livability Index

The Livability Index was calculated using each of the criteria, and the overall index is summarized in Table 6. As previously discussed, this study adopted the same scale as AARP. According to AARP [85], a score between 80–100 means that there are few, if any, challenges to living standards. The overall scores obtained for AUS was 0.87. The livability score obtained in this study was reported to be 87/100, which strongly suggests that there are few challenges at this university, as shown in Figure 10. In addition, all the scores for the individual criteria were above 0.80, which means that AUS is a highly livable campus based on the indicators identified in this study. The Index may inspire policymakers and professionals to go beyond their areas of expertise and understand the interconnected nature of livability.

Table 6. Results obtained for the Livability Index calculations for each criterion.

Criteria	Indicators	Scores
Environmental Quality	Amount of tree shading	0.84
	Amount of landscaping	
	Amount of outdoor lighting	
	Amount of public seating amenities	
	Street maintenance and cleanliness.	
Social and Cultural Elements	Places for informal leisurely gathering on campus (cafes, restaurants, etc.)	0.92
	Organized cultural events for different countries (art, fashion, food, etc.)	
	Organized social gatherings for special occasions	
	Socio-cultural competitions for kids (beauty contests, educational debates, etc.)	
Accessibility	Sidewalk physical condition (e.g., firm surface, slip-resistant, and free from cracks, bumps, free of litter, etc.)	0.88
	Parking spaces	
	Wayfinding signage	
	Campus connectivity to the surrounding areas Public transport around campus	
Safety	Crime rate	0.81
	Visibility of security officers	
	CCTV cameras	
	Threats from untamed pets	
	Gender harassment outdoors	
Amenities and Services	Healthcare services	0.86
	Facilities for special needs	
	Sports and recreation facilities	
	Emergency services on campus	
	Quality of technology (Wi-Fi, phone, cable, etc.)	
Housing	Various housing Choices	0.87
	Housing maintenance Services	
	Accessibility to residential areas (by foot)	
	Sense of privacy in the dwelling units	
	Quietness of residential complexes Neighborliness/friendliness of residents	
'Other' services	Places of worship (mosques, etc.)	0.91
	Banking services	
	Postal services	
	Veterinary services	
	Public transport to off campus places Services and utilities provided for building users	

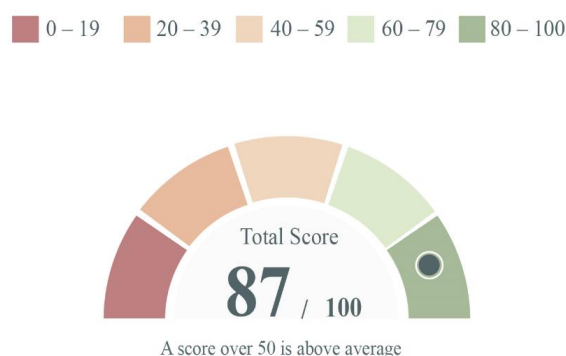


Figure 10. Livability Score of AUS.

4. Discussion

Building on the livability literature and on the comparisons of some studies that refer to universities as miniature cities [33], the present study proposes a set of criteria to compose a livability index for university campuses, using the American University of Sharjah (AUS) campus as a case study. The index obtained in this study serves as a reference/benchmark for similar and comparative studies of campus livability, where communities can add criteria or factors to the index, since it is an adaptable tool.

Consistently with other studies, the environmental criteria tend to score higher than other criteria in an index [50,52,86–88]. Gulwadi et al. [89] claims that the environmental quality of campuses (including greenness and landscaping) is associated with a better QOL for students specifically and other stakeholders in general. Lozano [90] found that the focus on the environmental quality dimension in universities may be due to the concept of sustainability having primarily environmental connotations. Traditionally, sustainability and livability have been linked to the environmental dimension. It may also be, as Dresner [91] posits, that there is a tendency to be more concerned with environmental issues rather than with social ones.

At AUS campus, the ‘Social and Cultural elements’ criterion had the lowest score in the proposed livability index. However, respondents were satisfied with the places for informal leisurely gatherings. This finding contradicts that for the University of Qatar [52], where many students argued about the lack of activities in general, which was the largest problem regarding socializing and the use of the space. This highlights the need for the university environment to have a widespread scope of activity. Aziz et al. [92] argue that universities should not be established simply to provide an academic curriculum, but more importantly, should be places to catalyze social interactions. However, AUS participants highlighted that they enjoyed parks and walking, and that the public realm of the AUS campus should be further developed to enhance social activities. Therefore, there is a need to enhance these measures at AUS to increase its livability.

In terms of transportation and accessibility to facilities and amenities, those two criteria were ranked highly at the AUS campus. However, the ‘Public transport around campus’ indicator had the lowest score. This finding is in line with the University of Dammam [50] and the University of Qatar [52] where results indicated that there was no well-established public transportation system on campus, and that respondents used their personal vehicles to go to the university and move around the campus. On the other hand, in universities outside the middle east, such as the University of Barcelona [93], public transport is the mode of transport most used by the university community to commute to and around the campus. This result is influenced by the availability of trains and trams in such countries. According to Adenle et al. [94], any university working towards sustainability and livability deals with the issue of transportation and accessibility. Some of the policies that promote sustainable mobility in universities include providing bus services and pedestrian walkways within the campus, supporting carpooling and bicycling among the university community, and raising the cost of on-campus parking. These policies have been studied and suggested for a number of universities in different case studies, such as

the University of Western Australia [95], the University of Coimbra in Portugal [96], the University of Bergamo in Italy [97], and Deakin University in Australia [98].

Safety issues on university campuses concern many individuals, including students, staff, and campus-based officials [99]. Campus safety is a complex topic encompassing the issues of violence, crime, and traffic incidents. All these incidents have indicated the complex nature and the importance of campus safety, including in universities in Malaysia [87], Turkey [86], Saudi Arabia [50], Qatar [52], USA [89], and China [100]. The usual recommendations to universities include training campus public safety and local law enforcement to react to critical events, employing communication systems to alert students, staff, faculty, and parents about emergencies on campus, making safety and security material accessible to the campus community, increasing the visibility of campus safety employees, and refining their communications with the campus community through several procedures, such as the use of foot patrol, and limiting access control to campus facilities [101,102]. At AUS, many of these measurements are applied, and they were ranked high by the respondents. It is worth noting that the respondents thought that their capacity to live comfortably was significantly impacted by built-in housing amenities, public amenities, and socio-psychological demands. This result demonstrates that everyone's ultimate goal is to have a safe and secure home that is surrounded by a strong family and neighborhood. Thus, it can be said that AUS housing is the type of home that ensures the residents quality of life at all phases of their lives, is simple to access and navigate within, and can accommodate residents' changing demands.

In light of the nature of this research and the concerns about data sources, this study faces some limitations. The limitations perceived in this study include:

- Inaccessibility and/or unavailability of the experts and/or key people who needed to be contacted by the researcher for surveys or interviews.
- Both quantitative and qualitative aspects utilized in this study are not conclusive. Adding additional aspects might assist in identifying other indicators of urban livability.
- The unavailability of some crucial secondary data suitable to be built into the index is another a limitation to the study. The majority of the primary data used in the study were gathered through sample surveys. The study's scope could be reduced by the lack of temporal data.
- Livability was contextualized from the perspective of AUS experts. There may be additional considerations for livability for the AUS campus context that could be investigated further with a wider range of stakeholders.
- Since this framework was created specifically for the AUS context, the project's findings might not be easily repeatable or transferable to other college campuses, due to diversity of the student community (where the UAE is composed largely of expats, and only a few of the locals reside on campus). However, with input from local stakeholders, this index and the techniques employed might be modified to be utilized at other universities.

The results of this study contribute to the existing body of knowledge by adding the current state of the art in relation to livability indicators within educational settings. Moreover, it contributes to the discourse in the field of livability theory by presenting latent (criteria) and observed variables (indicators) that propose key factors and their interrelationships so that livability can be addressed adequately.

This study proposed a holistic framework by collecting criteria derived from the literature, which were then evaluated and validated by experts, and which underpin the concept of livable campuses. This framework fills a current gap in the literature by offering academics and stakeholders a comprehensive review of the definition of a livable campus. This study also offers a contribution by investigating how important stakeholders perceive the livable campus requirements to be. This study thus paves the way for more research that will focus on stakeholders' interests in investing in the development of livable campuses and identify the crucial factors that matter to certain beneficiaries.

The results could be used to help AUS authorities understand what features need to be constructed to help promote a livable campus. In addition, other higher education institutions around the world could perform similar assessments to help them achieve a more livable campus

by working with all campus stakeholders to integrate livable factors into campus operations and research to create a more livable institution. It will require the university to aspire to the long-term goal of livability while addressing the daily needs that influence the university's impact on its residents. Moreover, institutions will need to identify both short-term and long-term goals that are SMART (specific, measurable, achievable, relevant, and time-bound). Before concrete design and planning recommendations can be supplied to urban designers and planners in the construction of university campuses, further research is required regarding the aspects that determine the occupancy trends in a diversity of campus design examples.

5. Conclusions

This study set out to propose a set of criteria to formulate a livability index for a university campus to achieve livability, using the American University of Sharjah (AUS) campus as a case study. An extensive literature search was used to obtain the best livability indicators using the Delphi method for a panel of experts and questionnaires for the residents. The proposed livability index considers seven dimensions of livability (Environmental Quality, Social and Cultural Elements, Accessibility, Safety, Amenities and Services, Housing, and Other Services) captured through 37 indicators. The proposed livability campus framework was created, where the insights from respondents and the results of the indicators allowed for the creation of a proposed campus design to enhance livability. Communities can add criteria or factors to the index, since it is an adaptable tool. One interesting finding that may be a subject for further research was that 96% of the respondents were non-Emiratis, while only 4% were Emiratis, which could be further investigated, as in the Emirati culture, the norm is that citizens live in their family compounds, so it was surprising that even 4% lived on campus. Consequently, this study suggested guidelines to aid policymakers to identify the aspects impacting university campuses and develop improved campus policies that ensure a thriving environment.

As previously mentioned, livability is a subjective measure, but it can include objective measures; hence, further research on livable campuses may quantify the importance of the prioritized criteria and their associated indicators within the study area through statistical models. There is also a need for a study with wider scope of livability among UAE universities, which might result in different findings and insights. Future research covering other universities could show whether student, faculty, and staff awareness and attitudes towards livability varies for different disciplines and between campuses to promote campus livability in UAE context and worldwide.

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Appendix A. Delphi Process to Arrive to the Final List of Indicators

Table A1. Delphi Process to Arrive to the Final List of Indicators.

Round 1 Original Indicators (Literature Review)	Round 2	Round 3
Environmental Quality Amount of landscaping Amount of lighting Amount of seating areas Street maintenance and cleanliness Variety of leisure services on campus Public cultural events for different countries Organized social gatherings for special occasions	Environmental Quality Amount of shading Amount of landscaping Amount of lighting Amount of seating areas Street maintenance and cleanliness Social and Cultural Elements Variety of services on campus Public cultural events for different countries Organized social gatherings for special occasions Socio-cultural competitions for kids	Environmental Quality Amount of tree shading Amount of landscaping Amount of outdoor lighting Amount of public seating amenities Street maintenance and cleanliness. Social and Cultural Elements Variety of entertaining services on campus Public cultural events for different countries (art, fashion, food, etc.) Organized social gatherings for special occasions Socio-cultural competitions for kids (beauty contests, educational debates, etc.)
Accessibility Accessibility Parking spaces Connectivity	Accessibility Sidewalk quality Accessibility for special needs Parking spaces Wayfinding signage Campus connectivity to the surroundings	Accessibility Sidewalk physical condition (e.g., firm surface, slip-resistant, and free from cracks, bumps, free of litter, etc.) Accessibility for special needs Parking spaces Wayfinding signage Campus connectivity to the surrounding areas Public transport around campus
Safety Crime rate Visibility of security officers CCTV cameras	Safety Crime rate Visibility of security officers CCTV cameras Threats from tamed and untamed pets	Safety Crime rate Visibility of security officers CCTV cameras Threats from tamed and untamed pets Gender harassment outdoors

Table A1. Cont.

Round 1 Original Indicators (Literature Review)	Round 2	Round 3
Amenities and Services Healthcare services Facilities for special needs Sports facilities	Amenities and Services Healthcare services Facilities for special needs Sports and recreation facilities Emergency services on campus	Amenities and Services Healthcare services Facilities for special needs Sports and recreation facilities Emergency services on campus Quality of technology (Wi-Fi, phone, cable, etc.)
Housing Housing maintenance Services Accessibility to residential areas Sense of privacy in the dwelling units Quietness of residential complexes	Housing Various housing Choices Housing maintenance Services Accessibility to residential areas Sense of privacy in the dwelling units Quietness of residential complexes	Housing Various housing Choices Housing maintenance Services Accessibility to residential areas (by foot) Sense of privacy in the dwelling units Quietness of residential complexes Neighborliness/friendliness of residents ‘Other’ Services Places of worship (mosques, etc.) Banking services Postal services Veterinary services Public transport to off campus places Services and utilities for building users

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